

Student Dashboard – Project Report

This report outlines the design, development, and insights of the Student Dashboard created using Power BI. The dashboard visualizes student data across multiple dimensions including demographics, academic performance, and engagement metrics.

Data Overview

The dataset includes the following key columns:

- `Roll No` (Primary Key)
- `Age`, `Age Group`
- `Gender`, `Country`, `Place Count`
- `Join Date`, `Department`
- `Attendance`, `GPA`, `Scholarship`
- `Extracurriculars`, `Cut Column` (cleaned)
- Additional engineered features for better segmentation and visualization

Exploratory Data Analysis (EDA)

Before building the dashboard, a thorough EDA was conducted:

- **Missing Values**: Imputed or removed based on context
- **Outliers**: Detected in GPA and attendance; handled using IQR method
- **Data Types**: Converted `Join Date` to datetime format
- **Feature Engineering**:
 - `Age Group`: Categorized into bins (e.g., 17–19, 20–22)
 - `Place Count`: Aggregated for geographic analysis
 - `Roll No`: Set as unique identifier

Dashboard Design Philosophy

The dashboard was designed to be:

- **Intuitive**: Easy to navigate and interpret
- **Insightful**: Highlights key trends and patterns
- **Interactive**: Filters and slicers for dynamic exploration

📊 Visualizations & Rationale

1. **Bar Chart – Department-wise Student Count**

- **Purpose**: Compare student distribution across departments
- **Insight**: Identifies departments with highest and lowest enrollment
- **Why Bar Chart**: Ideal for categorical comparison

2. **Pie Chart – Gender Distribution

- **Purpose**: Visualize gender ratio
- **Insight**: Reveals diversity metrics
- **Why Pie Chart**: Effective for showing proportions

3. **Geo Location Map – Country-wise Student Origin

- **Purpose**: Map student origins globally
- **Insight**: Highlights international reach
- **Why Map**: Adds geographic context and visual appeal

4. **Age Group Distribution

- **Purpose**: Segment students by age
- **Insight**: Most students fall within 18–22 age group
- **Why Column Chart**: Clear comparison of age bins

5. **Scholarship vs GPA – Scatter Plot

- **Purpose**: Analyze correlation between academic performance and financial aid
- **Insight**: High GPA students tend to receive scholarships
- **Why Scatter Plot**: Best for showing relationships

6. **Attendance Trends – Line Chart

- **Purpose**: Track attendance over time
- **Insight**: Detects dips and spikes in engagement
- **Why Line Chart**: Ideal for time-series data

7. **Extracurricular Participation – Stacked Bar Chart

- **Purpose**: Compare involvement across departments
- **Insight**: Some departments show higher participation
- **Why Stacked Bar**: Shows both total and breakdown

💡 Key Insights

- Departments with highest student enrollment are X and Y
- Gender ratio is approximately 60:40 (M:F)
- Students from over 10 countries are represented
- Scholarship allocation favors high-performing students
- Attendance dips during mid-semester periods
- Extracurriculars are most popular in Arts and Humanities

📌 Recommendations

- Increase outreach in underrepresented departments
- Review scholarship criteria for fairness and inclusivity
- Enhance engagement strategies during low-attendance periods
- Promote extracurriculars in STEM departments

🧠 Conclusion

The Student Dashboard provides a comprehensive view of student data, enabling stakeholders to make informed decisions. With interactive visuals and meaningful insights, it serves as a powerful tool for academic planning, diversity analysis, and student engagement.